Feedback Attention-Based Dense CNN for Hyperspectral Image Classification

Chunyan Yu, *Member, IEEE*, Rui Han, Meiping Song, *Member, IEEE*, Caiyu Liu, and Chein-I Chang, *Life Fellow, IEEE*

***Abstract*— Hyperspectral image classification (HSIC) methods based on convolutional neural network (CNN) continue to progress in recent years. However, high complexity, information redundancy, and inefficient description still are the main barriers to the current HSIC networks. To address the mentioned problems, we present a spatial-spectral dense CNN framework with a feedback attention mechanism called FADCNN for HSIC in this article. The proposed architecture assembles the spectral-spatial feature in a compact connection style to extract sufficient information independently with two separate dense CNN networks. Specifically, the feedback attention modules are developed for the first time to enhance the attention map with the semantic knowledge from the high-level layer of the dense model, and we strengthen the spatial attention module by considering multiscale spatial information. To further improve the computation efficiency and the discrimination of the feature representation, the band attention module is designed to emphasize the weight of the bands that participated in the classification training. Besides, the spatial-spectral features are integrated and mined intensely for better refinement in the feature mining network. The extensive experimental results on real hyperspectral images (HSI) demonstrate that the proposed FADCNN architecture has significant advantages compared with other state-of-the-art methods.**

***Index Terms*— Attention map, convolutional neural network (CNN), dense feature, hyperspectral image classification (HSIC), spatial feature extraction, spectral feature extraction.**

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Chunyan Yu, Rui Han, Meiping Song, and Caiyu Liu are with the Center for Hyperspectral Imaging in Remote Sensing (CHIRS), Information and Technology College, Dalian Maritime University, Dalian 116026, China (e-mail: yucy@dlmu.edu.cn; 269899266@qq.com; smping@163.com; liu_caiyu@163.com).

Chein-I Chang is with the Center of Hyperspectral Imaging in Remote Sensing (CHIRS), Information and Technology College, Dalian Maritime University, Dalian 116026, China, also with the Department of Computer Science and Information Engineering, National Yunlin University of Science and Technology, Douliu 64002, Taiwan, also with Remote Sensing Signal and Image Processing Laboratory, Department of Computer Science and Electrical Engineering, University of Maryland, Baltimore County, Baltimore, MD 21250 USA, and also with the Department of Computer Science and Information Management, Providence University, Taichung 02912, Taiwan (e-mail: cchang@umbc.edu).

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I. Introduction

HYPERSPECTRAL technology captures beneficial and accurate information of the landform with hundreds of subdivided spectral bands simultaneously, which plays a substantial role in many fields such as geological survey, crop monitoring, and national defense [1]–[3]. Hyperspectral image classification (HSIC) is a significant task in the remote sensing application area that assigns different regions with labels based on the multiple dimensional feature space [4]–[6]. In the last two decades, various strategies have been dedicated to implementing the HSIC. Since the spectral characteristic can natively reflect the signature of the different materials, one category of HSIC methods obtains the classification map in a pixelwise way, such as Fisher-related criterion [7] and statistical analysis [8]. This kind of approach yields unsatisfactory results due to the dimensional curse and the ignorance of the spatial information.

To further improve the performance of HSIC, another category of HSICs exploiting the integral effect on the spectral-spatial feature expression was proposed and proven to be successful. One type of existing approach increased the spatial information in the feature extraction stage, which is an effective way for HSIC. Zhao and Du [9] proposed a spectral-spatial feature-based classification (SSFC) framework to combine the spatial feature information with a local discriminant spectral embedding of the HSI. Benediktsson *et al.* [10] utilized various morphological decomposition methods to extract spatial spectrum features and successfully applied them to urban remote sensing image classification. Another type of approach accomplished spectral-spatial information classification by combining neighborhood spatial information, which performs classification after integrating features of the multiple adjacent pixels. Kang *et al.* [11] proposed the HSI feature fusion pattern cooperated by supporting vector machine (SVM) and the edge-preserving filtering (EPF). The methods [12]–[14] presented multiple linear manifolds to conduct the spatial-spectral feature characterization. Yu *et al.* [15] proposed a sparse representation for the characterization of the HSI features based on projection-based methods. In [16], an MASR model achieved excellent performance by utilizing a sparse policy adaptively to represent the HSI sample. Considering the difficulty of the complexity of the HSI data, the shallow-structured models mentioned above have a limitation on the practical features learning for classification.

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With the advancement of artificial intelligence and the development of deep learning, a wide range of HSIC methods based on CNN have attracted considerable attention to solving the nonlinear structure of the hyperspectral data [17]–[24]. The convolutional network of 1D CNN with multiple layers for HSIC [25] was designed to extract spectral features for the first time. Yu *et al.* [26] proposed a CNN architecture embedded with an extracted hashing feature to improve the accuracy of HSIC. In [27], a recurrent neural network (RNN) that exploited the following information of spectral correlation and band variability was proposed to deal with the spectral data. To construct a better representation of the classification feature, the spatial-spectral based 2-D CNN networks have made prosperous progress in recent years. In [28], a spatial transformation network was explored for CNN-based (STN) for HSIC to obtain the optimal input, which regularized the net with a designed DropBlock. Feng *et al.* [29] constructed a CNN architecture (MSLN-CNN) that considered multilayer spatial-spectral feature fusion for discriminative feature representation, and sample augmentation with local and nonlocal constraint was employed to overcome the overfitting problem of the HSIC. Gong *et al.* [30] proposed a multiscale convolution network (MS-CNNs) to extract discriminative features with the deep diversity-promoting metric. In [31], a spectral–spatial unified network (SSUN) with an end-to-end architecture was developed for HSIC, in which the spectral feature extraction, the spatial feature extraction, and classifier training were combined into a unified framework. Except for the 1-D CNN and 2-DCNN networks, the 3-D CNN approach for the spectral-spatial HSIC obtained the integrated spectral-spatial feature, which considers the inherent continuity of the 3-D HSI cube. In [32], a 3-D convolutional autoencoder (3D-CAE) network learned the spectral-spatial information of HSI. Roy *et al.* [33] proposed a hybrid spectral CNN (HybridSN) for HSIC, which consisted of a spectral-spatial 3D-CNN and a spatial 2D-CNN to joint spatial and spectral feature representation.

Meanwhile, with the development of the CNN architecture, a series of popular deep learning-based models have been applied in HSIC and achieved extraordinary success. In [34], a robust spectral-spatial HSI classifier was employed for feature fine-tuning based on generative adversarial networks (GANs). A cascaded model [35] was built to explore the complementary information by utilizing two RNN layers for HSIC. In [36], a novel squeeze multibias network (SMBN) was presented for HSIC, which developed a multibias module to increase the multibias information to the ReLu layer. Ma *et al.* [37] built a spectral-spatial capsule-based CNN framework to achieve better classification performance and reduce the complexity of the network design. In [38], a deep multilayer fusion dense network (MFDN) for HSIC was established to improve the performance. Recently, attention-based networks motivated by the squeeze-and-excitation scheme were built to exploit the information aggregation for HIS application. In [39], a spatial-spectral HSIC architecture was proposed by performing ResNet with a visual attention mechanism to enhance the quality of the feature. In [40], an adaptive residual convolutional neural network (ARCNN) was designed to mine discriminative features for HSIC. The residual blocks are responsible for 3-D convolutional layer connection through identity mapping. Wang *et al.* [41] introduced a spatial-spectral squeeze-and-excitation (SSSE) module that utilized squeeze operator and activation operations to refine feature maps. Besides, the popular models with embedded attention scheme can improve the classification performance. Mei *et al.* [42] proposed an attention-based bi-RNN network for spectral and spatial features extraction. In which the attention maps are calculated with a Tanh function and a sigmoid activation function. Moreover, a band attention module (BAM) was adopted to complete the band selection of HSI in [43]. The proposed BS approach was designed based on the assumption that the spectral band set can be reconstructed with a number of salient bands.

Although outstanding approaches have been made in the field of spectral-spatial HSIC method, it is still challenging to enhance discriminative features and improve efficiency in terms of accuracy pursuit. In this work, the feedback-attention-based dense CNN (FADCNN) is presented to characterize the spectral and spatial features for the HSIC. The purpose of the established model is able to build discriminative representation for the spectral and spatial characteristics, and enhance the robustness of the features with integrated learning in the way of dense concatenation. To fulfill the exactly above task, we construct three distinctive components in the proposed FADCNN architecture. First, to decrease the training complexity and time consumption, a BAM based on a fully connection network is designed, which produces the band weighted coefficient to constrain the bands for training. Second, the overwhelming dense features reveal the redundancy and overfitting problem, which leads to lower classification accuracy with more dense features. Recently, the attention mechanism is extensively applied on a range of tasks, including computer vision and natural language processing [45]–[47], which is motivated by human visual attention on the saliency regions of an object. Inspired by the fact that the attention mask can strengthen the representation discrimination, the spatial and spectral attention modules are constructed separately to derive competitive feature representation in this article. In which the high-level semantic information participates in the attention calculation in a feedback style, and the multiscale maps are organized to enhance the feature description. Last, to further extract the dense feature of HSI for classification, the feature mining (FM) module is built to combine the spatial and spectral features for incorporation, which intends to refine more accurate information and improve the classification accuracy.

The main contributions of this article are summarized as follows.

1) The presented HSIC architecture is collaborated by two-phase networks to estimate information robustly and effectively. Specifically, the proposed dense CNN framework that attributes to the first-phase network assembles the features in a compact connection style. The second-phase deep mining network reunites and furtherly refines the derived spatial-spectral features.
2) The proposed ADCNN model supplies two feedback attention modules for spectral and spatial attention-based

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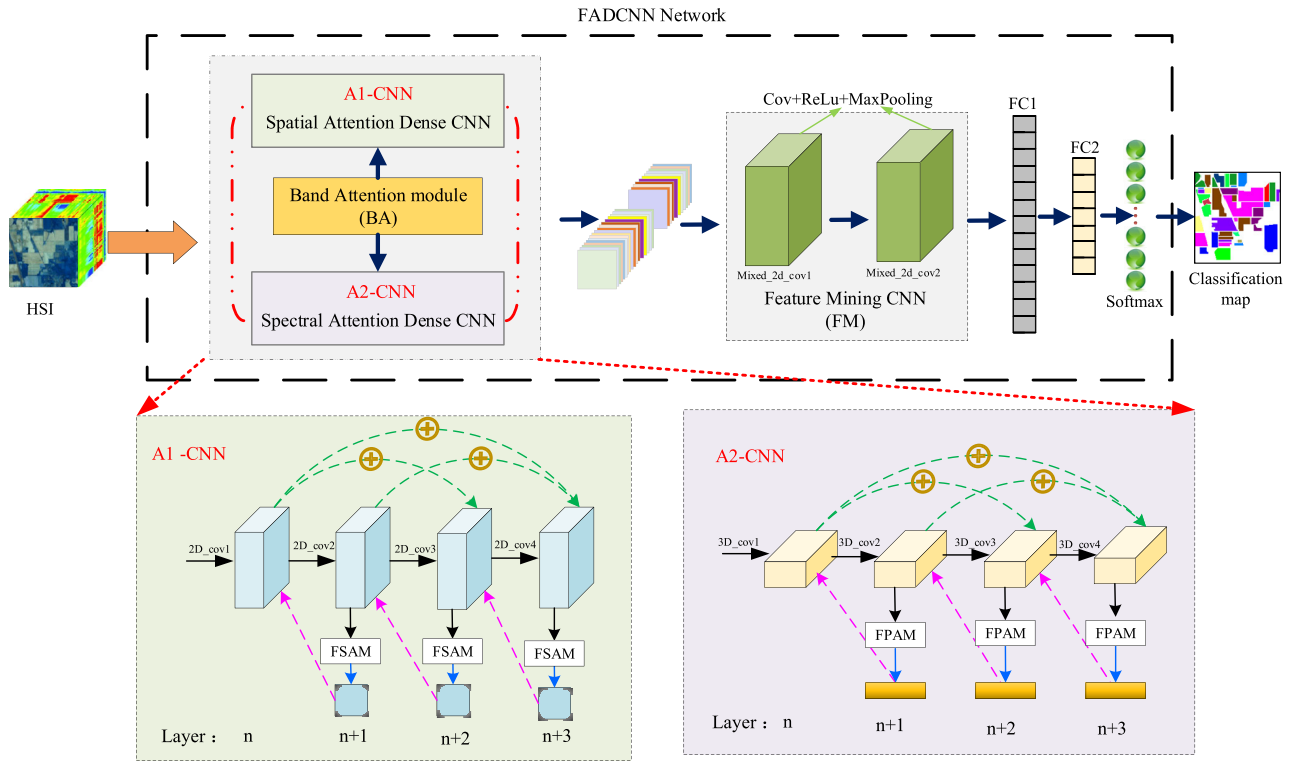


Fig. 1. Hierarchical structure diagram of the proposed FADCNN network. The top model illustrates the overview of our model that is composed of three distinctive components, the BA component for band coefficient inferring, the spatial and spectral attention dense CNN for feature extraction, and the FM block for feature refinement. (Bottom-left) A1-CNN model and (Bottom-right) A2-CNN model demonstrate the details of the dense networks with a feedback attention mechanism.

maps calculation, which acquire attention masks in a feedback way from the subsequent convolution block. To the best of our knowledge, it is the first attempt to compute the attention maps in the feedback style. Unlike the previous research, the attention modules are designed with two schemes to enhance the saliency ability expression. On one aspect, semantic information from the higher layer block is involved with attention map computation. On the other aspect, the multiscale context is explored for spatial attention to strengthen the feature description.

3) As a critical part of the model, the proposed BA component is expected to infer the attention coefficient of the band. Only the determined attention-based bands participate in sequential training with the given priority, which guarantees that the amount of training data reduces dramatically. The proposed block supplied in an unsupervised fashion is easily embedded in other CNN networks to reduce training consumption.

The remaining part of this article is structured as follows. The details of the proposed feedback dense CNN for HSIC is described in Section II. Section III illustrated the experimental results and analysis, and conclusions are drawn in Section IV.

II. Proposed Approach

The flowchart of the proposed FADCNN framework is illustrated in Fig. 1. It mainly consists of distinctive components: the BAM, the FM CNN block, and attention aided dense networks, including a dense spatial subnetwork denoted as A1-CNN and a dense spectral subnetwork recorded as A2-CNN. In particular, the BA module based on a fully connection network is employed to emphasize the effectiveness of the training band and decrease training consumption. The dense feature with attention mechanism in the A1-CNN and A2-CNN are designed to better characterize the features for description, and the extracted dense features are feed into the FM block to collect the discriminative information with further refinement.

To accomplish the proposed model, we first adopt the BA module as a powerful strategy to obtain the attention bands of the original cube. A1-CNN and A2-CNN networks are employed to extract the dense spatial and spectral features with feed attention block denoted as FSAM and FPAM independently during the training procedure. Next, the compact features are combined and further mined through the FM block. Lastly, the discriminative features are assembled by two fully connection layer and activated by the soft-max function, and the training loss is calculated by the cross-entropy function. More details of each module are described in the following sections.

A. BA Component

HSI cube is composed of abundant bands with inevitable correlation, to increase the effect of the different band for the data training, the BA implementation is adopted in this section. Noticed that the band closest to others implied the

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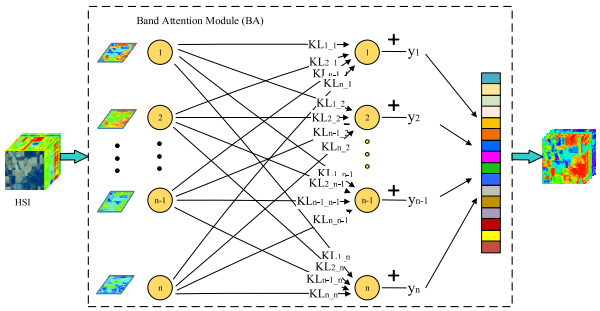


Fig. 2. Structure of the BA approach.

most discriminative information, in this article, a feed-forward fully connection network with two layers is employed to compute the similarity of the bands regarding the choice of the informative spectrum. Fig. 2 shows the flowchart of the proposed BA approach.

With the initial band similarity calculated by Kullback–Leibler (KL) divergence as input to promote the distribution of band, the output of each node is utilized to measure the policy of the BA. Let b_i and b_j be the two bands of the HSI, the divergence of the ith band and the jth band recorded as $KL(i_j)$, which is calculated by the following equation:

$$KL(i, j) = \sum_{k=1}^{n} b_{ik} \log \frac{b_{ik}}{b_{jk}} \tag{1}$$

where b_{ik} denotes the kth pixel in the ith band, n represents the number of pixels of the HSI. The coefficient of the ith band is denoted as y_i in terms of the following equation:

$$y_i = \sum_{k=1}^{n} KL(i, n). \tag{2}$$

After the forward propagation with the weight of 1 and the bias of 0, the output column vector is denoted as $ba = (y_1, y_2, \ldots, y_N)^T$, which represents the influence of full band for the HSI. Let X_i denotes the ith band of HSI, after the BA module, the critical band influence of individual band is counted in the training phase, and the training input $X = \{X_1, \ldots X_i, \ldots\}$ transforms to the compacted representation by

$$X \rightarrow X \otimes ba \tag{3}$$

where $\otimes$ denotes elementwise multiplication.

B. Spatial-Spectral Dense Network

As illustrated in the A1-CNN and A2-CNN in Fig. 1, the proposed spatial-spectral dense network is utilized to extract the dense feature in spatial and spectral levels independently. The spatial feature $F_1(X_n^{A1})$ performed convolution is formulated by

$$F_1\left(X_n^{A1}\right) = \sigma\left(X_n^{A1} \Theta w^{2D} + b^{2D}\right) \tag{4}$$

where w^{2D} and b^{2D} are the weights and bias of the network, Θ refers to the 2-D convolution operator. Here, the nonlinear $\sigma(\cdot)$ denotes the ReLu function.

We utilize the dense block to enhance the retaining and effectiveness of the spatial features, which concatenate the output of all previous layers in a feed-forward style. A dense operation is defined as

$$X_n^{A1} = \sum_{p=1}^{n-1} \left(X_p^{A1}\right) \tag{5}$$

where X_n^{A1} represents the spatial input at the nth layer. In this way, the input of the current layer is not only related to the adjacent layer but also the other previous layers, which can explore refined features than the traditional implementation.

A 3-D CNN block is designed to complete spectral information extraction. In this section, the training sample for HSI is denoted as a 4-D tensor $X \in R_1^{n_h \times n_w \times l_h \times k_c}$, where $n_h \times n_w \times l_h$ is the size of the training sample, $k_c = 1$ is the number of the channel, respectively. In this article, we only capture the refined spectral feature through 3-D CNN denoted as $F_2(X_n^{A2})$ by the 3-D filter with the size $1 \times 1 \times k$ by the following equation:

$$F_2\left(X_n^{A2}\right) = \sigma\left(X_n^{A2} \odot w^{3D} + b^{3D}\right) \tag{6}$$

where w^{3D} and b^{3D} are the weight and bias of the filters, the symbol $\odot$ represents the 3-D convolution operator.

Similar to the implementation of the spatial feature enhancement, the dense block is utilized to concatenate the spectral feature of all previous layers too. Correspondingly, a dense operation for spectral information is expressed as

$$X_n^{A2} = \sum_{p=1}^{n-1} \left(X_p^{A2}\right) \tag{7}$$

where X_n^{A2} represents the spectral input at the nth layer. In a similar way, the learning procedure of spectral data involved the information from all the previous layers to enhance the feature description.

C. Feedback Attention Modules

The attention scheme is significant for strengthening the discriminant feature of the dense CNN. Inspired by the fact that deep layers build better contextual features than the shallow layer in CNNs, we develop an attention mechanism to predict semantic attention information. The attention mask of the sequential-layer feature map is calculated with the current-layer feature in a feedback style. Besides, the spatial attention map calibrates spatialwise information in a hierarchical mechanism by exploring more powerful features. In specific, the feedback spatial attention module (FSAM) and spectral attention module (FPAM) blocks are described in detail in the following parts.

As demonstrated in Fig. 3(a), we build the FSAM block by exploiting the feature attention with multiscale information confusion for spatial feature refinement. Given an intermediate feature map $F_1(X_{n+1}^{A1}) \in R^{c \times H \times W}$, where c represents channel number, H $\times$ W represents the feature size. A max-pooling and an average pooling operation are firstly applied to generate two maps. Then the map denoted as Y_{n+1}^{A1} is acquired by the pixelwise addition, which balances the local information and the global invariance of the $F_1(X_{n+1}^{A1})$. Next, the multiscale features of Y_{n+1}^{A1} are obtained as Y_1, Y_2, Y_3 by downsampling,

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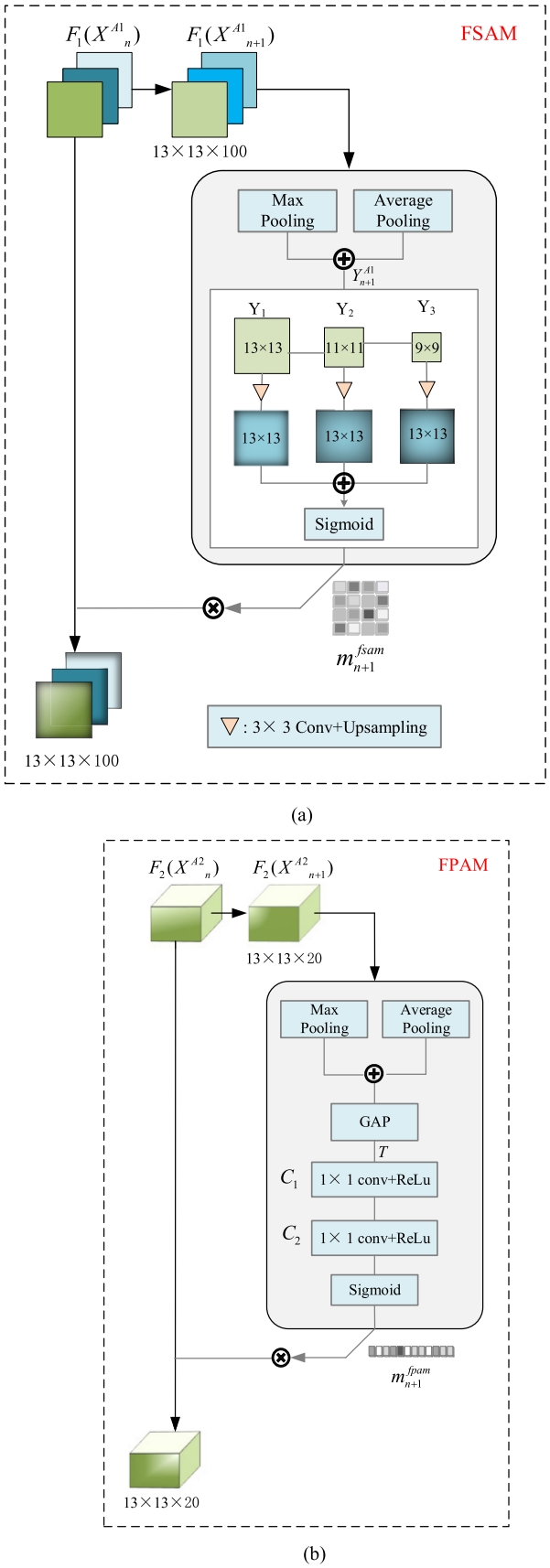


Fig. 3. Attention scheme design. (a) FSAM module. (b) FPAM module.

in which the size of three maps is 13×13, 11×11, and 9×9, respectively. The attention mask represented as $m_{\text{fsam}}^{\text{fsam}-i}$ of Y_i is produced by a 1×1 convolutional layer, and the upsampling operation is applied to rescale the aggregated attention map. Afterward, the fusion attention matrix of $F_1(X_{n+1}^{A1})$ recorded as m_{n+1}^{fsam} is calculated by $m_{n+1}^{\text{fsam}} = m_{n+1}^{\text{fsam}-1} \oplus m_{n+1}^{\text{fsam}-2} \oplus m_{n+1}^{\text{fsam}-3}$. Here, the symbol $\oplus$ denotes elementwise addition. Furthermore, a simple gating mechanism is used with the sigmoid function to complete the normalization of the attention map

$$m_{n+1}^{\text{fsam}} \leftarrow \delta\left(m_{n+1}^{\text{fsam}}\right) \tag{8}$$

where $\delta(\cdot)$ is the sigmoid function. FSAM perform attention mechanism combines high layer-level features for generating better attention prediction, and after the feed of the feature map of the n*th* layer into the FSAM block, a new refined spatial feature is created as

$$X_n^{A1\prime} = X_n^{A1} \otimes m_{n+1}^{\text{fsam}}. \tag{9}$$

Correspondingly, (5) is formalized as

$$X_n^{A1} = \sum_{p=1}^{n-1}\left(X_p^{A1\prime}\right). \tag{10}$$

Similar to the FSAM, the FPAM is constructed as illustrated in Fig. 3(b). The backbone network for FPAM is composed of one global average pooling and two convolution layers. First, a max-pooling and an average pooling operation are applied to generate the feature Y_{n+1}^{A2}. The global average pooling is used to create statistics of the channels, which is denoted as vector T. Next, the feature T is feed to two 1×1 convolutional layers, and a sigmoid activate function is applied to drive the normalization of the spectral attention map m_{n+1}^{fpam} by the following formula:

$$m_{n+1}^{\text{fpam}} = \delta(C_2(C_1(T))) \tag{11}$$

where C_1 and C_2 are two consequent 1×1 convolution layer with ReLu activation. Afterward, the spectral feature map of the nth layer is computed as

$$X_n^{A2\prime} = X_n^{A2} \otimes m_{n+1}^{\text{fpam}}. \tag{12}$$

Relevantly, (7) turns into

$$X_n^{A2} = \sum_{p=1}^{n-1}\left(X_p^{A2\prime}\right). \tag{13}$$

D. FM Block

The key to the HSIC framework is to incorporate the spectral and spatial features jointly to improve the performance of the proposed framework. In this section, the FM module is designed for the implementation of the fusion and keep mining of the dense spectral-spatial features. In specific, the features obtained from A1-CNN and A2-CNN are proceeded to construct a fusion formation. The procedure is represented as follows:

$$F = \sum_{l=1}^{L} F_1(X_l) \circ F_2(X_l) \tag{14}$$

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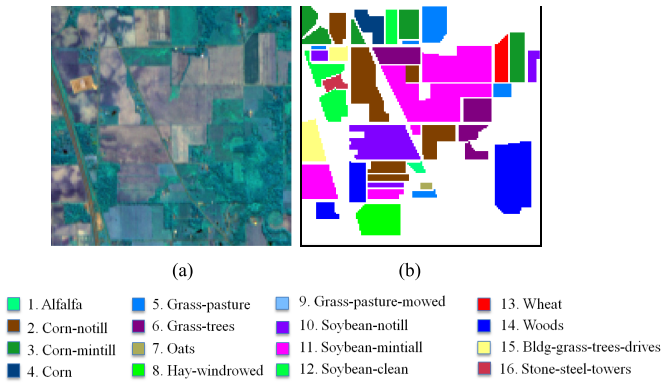


Fig. 4. Data of Purdue Indiana Indian pines scene. (a) False-color image (band 25, 36, 98). (b) Ground truth image.

where the symbol of $\circ$ denotes the concatenate operation, L is the number of convolution layers, and $F_1(\cdot)$ and $F_2(\cdot)$ describe the features obtained from A1-CNN and A2-CNN.

In addition, based on the combined feature F, a series of subsequent operations are adopted to refine the discrimination of the spatial-spectral feature in the FM network. Precisely, the FM network in this article for FM consists of two convolution blocks denoted as Mixed_2d_cov1 and Mixed_2d_cov2, and each block is composed of one convolution layer, a ReLu activation layer, and one max-pooling layer operation. The convolution layer mines the merged features of spatial-spectral characteristics jointly. The ReLu function is responsible for the perceptron ability of the nonlinear feature, and the max-pooling layer enlarges the receptive region of the refined features. Eventually, the optimized features are fed into two fully-connection layers named FC1 and FC2. The in-depth features are converted to a 1-D vector, and the softmax classifiers complete the classification mapping sequentially.

III. Experiment and Result Analysis

A. Data Description

In this section, we adopted three kinds of extensively used data sets to evaluate the performance of our proposed model. The first one is the Purdue Indiana Indian pines scene, which was captured over the Indian Pines test site in North-Western Indiana. The resolution of the data is 145 $\times$ 145, and it contains 200 spectral bands, the Purdue Indian Pines Scene consists of 16 classes of different plants, a false-color image, and the ground truth image are shown in Fig. 4(a) and (b).

The second data set is called the Kennedy Space Center (KSC) data, which was collected from the KSC located in Florida. The size of the data is 512 $\times$ 614 with 176 bands after water absorption bands removing. The false-color image and the ground truth image cover 13 types of swamp are shown in Fig. 5(a) and (b), respectively.

The last data set is the University of Pavia, which is acquired by the ROSIS-03 satellite sensor. This scene contains 610 $\times$ 340 pixels with a spatial resolution of 1.3 m and the spectral band number is 103. Fig. 6(a) shows a false-color composite image and the ground truth image with nine labeled classes is illustrated in Fig. 6(b).

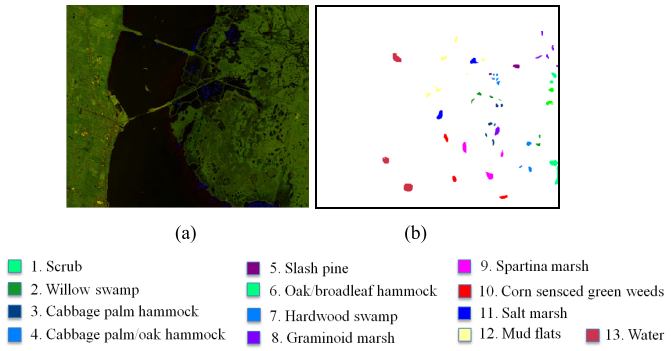


Fig. 5. Data of KSC. (a) False-color image (band 35, 76, 98). (b) Ground truth image.

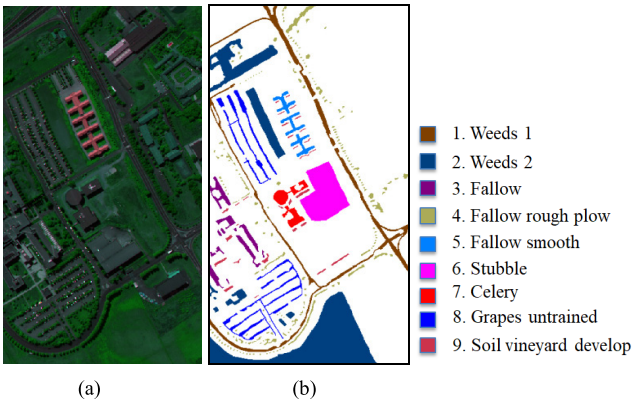


Fig. 6. Data of the University of Pavia. (a) False-color image (band 16, 75, 47). (b) Ground truth image.

B. Experimental Configuration

In order to explore the performance of the proposed FADCNN network for HSIC, a series of other popular models were exploited for comparison, including EPF [11], ITCIMC [48], MFASR [14], CNNDH [26], Bi-RNN [49], ResNet34 [50]. Besides, we also compare the proposed method with the PCA approach instead of the BA block on the three data sets. Here, the framework with the PCA with three and five primary components instead of the BA module is denoted as PCA3-F and PCA5-F, the FADCNN network without BA block is called the FADC method. The simplified explanation and implemented configuration of the compared models are listed as follows.

1) *EPF:* This method is a very classical spectral-spatial HSIC approach that utilizes EPF to build feature representation of HSIs for the first time. We applied it with the default parameters in this article, and please refer to [11] for the implementation details.
2) *ITCIMC:* This approach performed HSIC in an iterative feedback way, which captured and combined spatial contextual information to improve the classification precision. The input of this model is the spectral signature of the specific class computed with the label, σ used in the Gaussian filter is set to 0.5 with a window size of 5, and the stopping threshold is 0.99.

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TABLE I

HYPERPARAMETER SETTING OF THE PROPOSED MODEL

Module	Layer setting	Input size	Kernel size	Output size
A1-CNN	2D_cov1	13×13×u	3×3×100, stride=[1,1]	13×13×100
	2D_cov2	13×13×100	3×3×100, stride=[1,1]	13×13×100
	2D_cov3	13×13×200	3×3×100, stride=[1,1]	13×13×100
	2D_cov4	13×13×300	3×3×150, stride=[1,1]	13×13×150
	FSAM	–	–	–
A2-CNN	3D_cov1	13×13×u	1×1×3×1,stride=[1,1,1]	13×13×u
	3D_cov2	13×13×u	1×1×3×1,stride=[1,1,1]	13×13×u
	3D_cov3	13×13×2u	1×1×3×1,stride=[1,1,1]	13×13×2u
	3D_cov4	13×13×4u	1×1×3×1,stride=[1,1,1]	13×13×4u
	FPAM	–	–	–
FM	Mixed_2d_cov1	13×13×(450+8u)	3×3×150,stride=[1,1]	13×13×150
	Mixed_2d_cov2	13×13×150	3×3×200,stride=[1,1]	13×13×200
FC1	–	33800×1	1000	1000×1
FC2	–	1000×1	100	100×1
Softmax	–	100×1	–	N_C×1

3) *MFASR:* It supplied a sparse feature representation for HSI data based on multiple features extracted with different spatial-based filters. In our approach, we utilize four kinds of features with a sparsity level of 3.
4) *CNNDH:* The structure of the CNNDH is constructed the same as [26], which is composed of two convolutional layers, one deconvolution layer, and one embedded hash semantic feature extraction module. For a fair comparison, we accomplished CNNDH with the same hyperparameters shown in [26] except for the data augmentation phase.
5) *Bi-RNN:* The bidirectional RNN is accomplished by training simultaneously in a positive and negative time direction. In the experiment, we collected spatial patches with a size of 13 × 13 instead of spectrum information as a training sample. There are three hidden layers in our implementation, and each layer has 60 nodes. The other set includes the batch size of 20, iteration of 5000, and the learning rate of 1e-4.
6) *ResNet34:* We employed a baseline 34-layer ResNets for HSIC. The implemented configuration of each residual block is utilized with the default parameters in [50], we fix the number of iterations as 5000, and the learning rate of 1e-4.

The hyperparameters setting in terms of the modules of the FADCNN are listed in Table I. Here u represents the attention-based band number, and N_c is the number of the classes of the specific data set. According to the VD algorithm [51], the numbers of attention-based bands of the Purdue Indiana Indian pines scene and the KSC are 18 and 21 for the proposed model, respectively, and for the Pavia data, we set the attention-based number of 14. In the flowing experiments, we set the attention-based y_i with the priority of 1 and the other y_i with the priority of 0. It can be seen that the amount of training data of the network model has been dramatically reduced.

In the following experiments, we utilize the overall accuracy (OA) and average accuracy (AA) as objective criteria to evaluate the performance of all the compared methods. Each execution of all the networks has been repeated five times, and the OA calculation is reported in the form of mean ± standard deviation. The implementation of all the models has been accomplished in the platform of TensorFlow, the environment of the running is RTX 2080 Ti CPU and 32 G RAM. The batch size is initialized to 100, the proposed networks are trained for 5000 iterations, and the learning rate is 0.001. The adopted optimizer is Adam, the dropout rate of the three HSI data is fixed as 0.1, and the image size of the training sample is 13 × 13. Moreover, Tables II, III, and IV listed the number of samples for the training and testing sets of the three HSI data.

TABLE II

NUMBER OF SAMPLES IN THE TRAINING AND TESTING SET OF THE PURDUE INDIANA INDIAN PINES SCENE (10%)

Class No.	Class Name	Training	Testing
1	Alfalfa	5	41
2	Corn-notill	143	1285
3	Corn-mintill	83	747
4	Corn	24	213
5	Grass-pasture	49	434
6	Grass-trees	73	657
7	Grass-pasture-mowed	3	25
8	Hay-windrowed	48	430
9	Oats	2	18
10	Soybean-notill	98	874
11	Soybean-mintill	246	2209
12	Soybean-clean	60	533
13	Wheat	21	184
14	Woods	127	1138
15	Buildings-Grass-Trees-Drives	39	347
16	Stone-Steel-Towers	10	83

TABLE III

NUMBER OF SAMPLES IN THE TRAINING AND TESTING SET OF THE KSC (10%)

Class No.	Class Name	Training	Testing
1	Scrub	77	684
2	Willow swamp	25	218
3	Cabbage palm hammock	26	230
4	Cabbage palm/oak hammock	26	226
5	Slash pine	17	144
6	Oak/broadleaf hammock	23	206
7	Hardwood swamp	11	94
8	Graminoid marsh	44	387
9	Spartina marsh	52	468
10	Cattail marsh	41	363
11	Salt marsh	42	377
12	Mud flats	51	452
13	Water	93	834

C. Results and Analysis

In the experiment for the Purdue Indiana Indian pines scene, 10% of the training samples were randomly selected, and the input data size after BA is 13 × 13 × 18. The classification results of OA with the compared models are illustrated in Table V, and the colorful result maps are demonstrated in Fig. 7. As shown in Table V, the FADCNN achieved the highest OA of 97.95%. It can be analyzed that the dense block has a certain positive effect on the network model. Moreover, we observe that the performance of FADCNN is better than FADC of OA (95.71%), which can be noted that the BA method is essential and beneficial for the proposed architecture.

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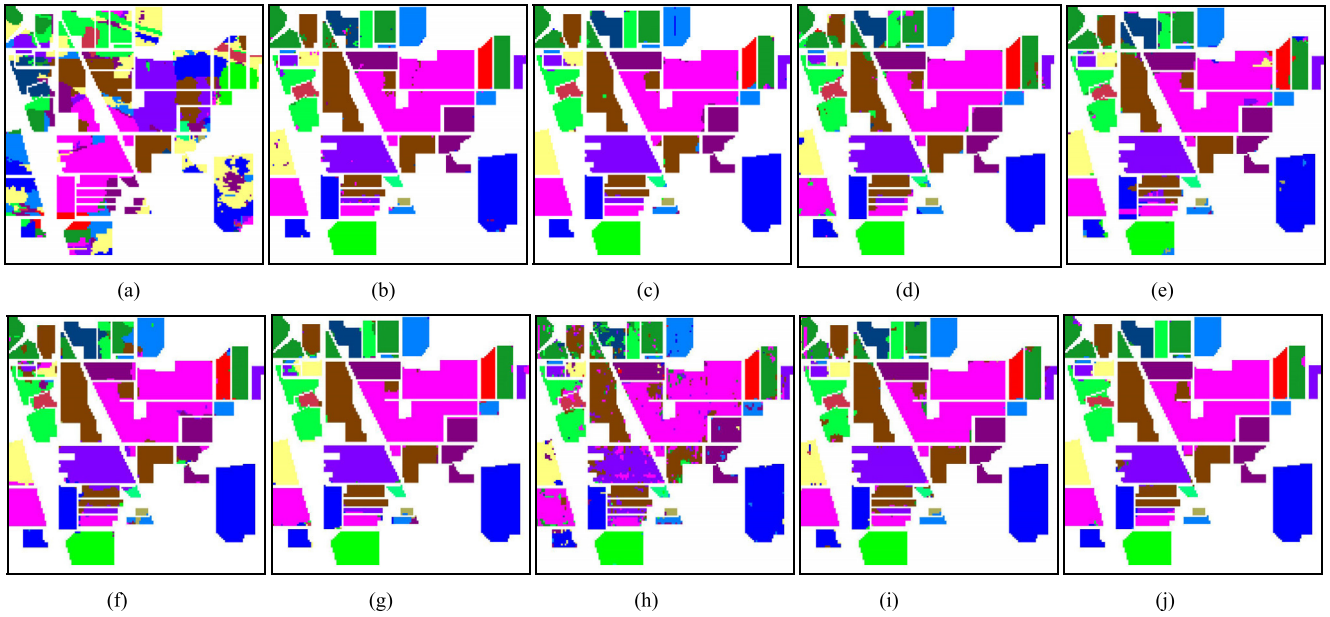


Fig. 7. Classification maps of the Purdue Indiana Indian Pines Scene with compared methods. (a) EPF. (b) ITCIMC. (c) MFASR. (d) CNNDH. (e) PCA3-F. (f) PCA5-F. (g) FADC. (h) Bi-RNN. (i) ResNet34. (j) FADCNN.

TABLE IV
NUMBER OF SAMPLES IN THE TRAINING AND TESTING SET OF UNIVERSITY OF PAVIA (5%)

Class No.	Class Name	Training	Testing
1	Asphalt	332	6299
2	Meadows	933	17716
3	Gravel	105	1994
4	Trees	154	2910
5	Painted metal sheets	68	1277
6	Bare Soil	252	4777
7	Bitumen	67	1263
8	Self-Blocking Bricks	185	3497
9	Shadows	48	899

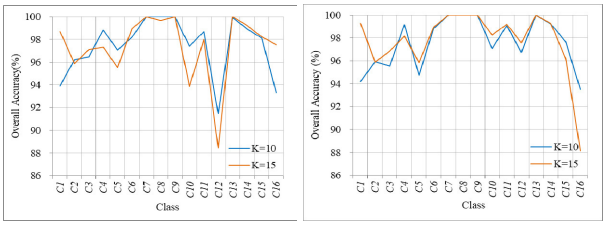


Fig. 8. Accuracy value of the Purdue Indiana Indian Pines Scene with FADCNN model when the number of training samples for classes 1, 7, 9 are fixed to 10 and 15. We report the average results of 5 experiments. (a) Randomly selection strategy. (b) Preserved selection strategy.

Compared with the MFASR, the FADCNN model reports the AA of 93.96%, which is lower than the value of 97.20%. The reason is that the number of training samples of classes 1, 7, and 9 is very few in the proposed network. In terms of this issue, we also conduct the experiment with the training number (K) fixed as 10 (first condition) and 15 (second condition) for classes 1, 7, and 9. In this experiment, the samples for other categories are randomly selected with a radio of 10%. The classification results of each class are demonstrated in Fig. 8(a). As expected, the OAs of class 1 are 93.91% with $K = 10$ and 98.70% with $K = 15$, respectively. For classes 7 and 9, the OAs reaches 100% when K equals 10 or 15. The detailed OA of each class show qualitative comparisons with different circumstances. In which classes 2, 4, 5, 8, 10, 11, and 12 are obtained better performance in the first condition, while classes 3, 6, 13, 14, 15, and 16 have higher OAs in the second situation. Besides, it should be noted that the experiment in the first condition achieves less realistic performance (97.55%) compared to the second condition (96.71%). The reason is that the samples of each other category are chosen randomly, which indicates the samples for any other class may have more discriminate features in the first condition than the second condition. To confirm this stated opinion, we have redone an experiment with our FADCNN model on Purdue. In this case, the strategy of the sample selection for the two situations is not totally random. We preserve the samples in the first condition, only increased five more random samples in classes 1, 7, and 9. The classification accuracy of each class is shown in the polylines in Fig. 8(b). As can be seen from the illustration, the performance of each category in the style of preserved sample selection has similar observation with the indication in Fig. 8(a), expectedly. Not all OAs in the second condition is significantly superior to the ones in the first situation. Meanwhile, the OAs in this conduction is ideally satisfactory, which are 97.87% and 98.02% for the case of $K = 10$ and $K = 15$, respectively.

The training sample percentage of the KSC is set to 10%, the sample size of the data set after the BAM was $13 \times 13 \times 21$. Fig. 9 illustrated the classification maps of the model. As shown in Table VI, it can be seen that the FADCNN model achieved the best OA of 99.10% higher than Bi-RNN (89.97%) and ResNet34 (90.94%). In comparison,

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TABLE V
OA FROM THE CLASSIFICATION RESULTS WITH ALL THE COMPARED METHODS OF THE PURDUE INDIANA INDIAN PINES SCENE

Class P_{OA}%	EPF	ITCIMC	MFASR	CNNDH	PCA3-F	PCA5-F	FADC	Bi-RNN	ResNet34	FADCNN
1	**100.00**	95.65	**100.00**	60.43±2.83	99.13±1.95	82.17±2.83	94.35±8.22	79.46±11.08	80.00±10.12	88.26±7.73
2	83.68	96.01	96.58	92.06±1.05	92.87±1.08	95.00±1.39	95.45±2.34	79.37±3.89	94.26±3.56	**98.04±0.27**
3	72.17	96.99	97.59	88.05±2.01	87.45±2.04	86.65±2.79	**99.06±0.65**	82.89±3.77	89.61±3.94	97.04±1.50
4	**99.58**	98.73	89.67	95.69±0.76	97.72±2.34	98.90±1.83	98.65±0.63	66.95±4.38	91.45±7.60	96.03±3.21
5	94.20	89.44	97.93	95.65±1.39	92.71±2.28	88.36±2.77	93.46±3.27	89.54±1.01	97.13±2.32	**99.34±0.44**
6	**99.59**	97.12	98.63	98.82±0.60	96.03±2.63	89.92±1.00	98.96±0.96	94.04±1.09	98.29±3.22	99.48±0.48
7	96.43	**100.00**	96.00	87.86±7.41	83.57±7.83	94.29±5.42	82.86±12.47	79.09±13.09	71.20±8.67	75.72±20.50
8	**100.00**	98.78	**100.00**	**100.00±0.00**	93.77±1.72	98.24±1.47	99.75±0.56	98.48±1.50	**100.00±0.00**	**100.00±0.00**
9	95.00	**100.00**	94.44	93.00±2.74	57.00±4.47	87.00±11.51	64.00±15.57	70.00±10.27	48.78±8.26	78.00±25.42
10	87.55	93.93	**97.71**	94.98±0.68	95.47±1.38	93.64±0.91	95.51±2.52	79.07±2.27	94.26±2.93	96.50±0.89
11	88.47	94.70	**99.00**	96.67±0.50	96.14±0.73	97.66±0.59	98.85±0.15	89.68±0.99	97.77±0.99	98.49±0.69
12	**97.64**	95.45	95.31	92.62±1.58	91.16±0.91	90.96±4.67	96.42±1.62	81.98±3.91	90.90±0.89	95.45±1.77
13	99.51	98.54	99.46	99.41±0.22	93.95±2.35	93.17±1.54	**100.00±0.00**	95.73±3.14	98.38±1.27	99.02±1.02
14	98.74	93.52	99.03	99.41±0.23	96.93±2.09	99.40±0.47	96.69±5.76	96.48±0.45	98.63±1.35	**99.48±0.66**
15	96.89	90.67	95.68	94.04±1.52	98.13±0.53	85.23±2.27	98.19±1.48	87.38±1.23	97.52±1.41	**99.74±0.16**
16	**100.00**	98.92	97.59	98.92±0.76	85.38±2.59	87.31±4.95	89.89±10.50	78.92±3.11	96.87±1.38	82.80±9.40
OA	90.78	95.09	97.83	95.29±0.15	94.32±0.47	94.16±0.31	97.25±0.83	87.08±1.51	95.71±0.86	**97.95±0.15**
AA	94.13	96.09	**97.20**	92.98±1.52	91.09±2.31	91.74±2.90	93.88±4.17	84.32±4.07	90.32±3.62	93.96±4.63

the FADC yields an OA of only 93.95%. The reason is that the KSC data set has a relatively sparse characteristic of the feature distribution, so that the traditional network generally has more difficulties in the representation of the spectral-spatial feature. While with the BA operation, the proposed model achieved the highest performance compared to the other state-of-the-art methods, which certifies the proposed FADCNN has the effectiveness with the better feature description. Besides, it needs to be noted that the PCA-based proposed models yielded the worst performance with OA of 38.78% for PCA3-F, and 44.47% for PCA5-F, respectively, which indicates that the representation of the primary components has terrible influence in the dense feature extraction block for the HSI data with the characteristic of high sparsity. In contrast, our approach with the BA module has better robustness for the sparsity.

The last experiment is performed at the University of Pavia. The data size is transformed into $13 \times 13 \times 14$ after the BA block, and the training ratio is set to 5%. The colorful classification maps are illustrated in Fig. 10, and the OAs are listed in Table VII. We can see that the OAs generated with PCA3-F, PCA5-F, and FADC are 97.76%, 98.13%, and 98.92%. Our proposed model can further increase the performance to 99.19% by incorporating the BA block. In similarity, the AA values are 96.91%, 96.99%, 98.26% for the PCA3-F, PCA5-F, and FADC model, respectively. As can be observed, the proposed FADCNN has a relatively stable and balanced classification effect for each class and generates the maximum AA value (98.49%). All the experiments conducted on the three data sets demonstrated that the FADCNN model reflects the excellence and robustness of HSIC.

Furthermore, we conducted a series of experiments with the same sample number for each class. In this occasion, the sample number varies from {5, 10, 15, 20, 30, 40, 50}. It needs to be noted that the strategy of sample selection is different from the random style. Specifically, the chosen samples for the last time are preserved, and only five more samples of each class are randomly increased for subsequent training. Accordingly, all the classification result maps are exhibited in Figs. 11–13 for Purdue, KSC, and Pavia, respectively. Besides, the OAs of each class are intuitively displayed in the charts of Fig. 14. As demonstrated in the diagrams, the FADCNN method produces higher OA with more samples involved in the feature extraction. Specifically, the OA of class 3 of the Purdue data is merely 14.54% with $K = 5$, and it increases to 91.97% with $K = 50$. For KSC, the approach with $K = 5$ gains not satisfactory OAs for classes 3, 4, 8 of 36.59%, 48.41%, and 41.40%, while the performance improves sharply with the addition of the K. In similarity, the performance of class 3 is ineffective with the OA of 24% for the Pavia. At the same time, the OAs progress significantly with a larger set of K.

D. Ablation Study

To demonstrate the contribution of different modules of the proposed approach to the ultimate classification result, we carry out the ablation study on the three data sets with the baseline network of A1 + A2. Individually, the results of the experiments are listed in Table VIII.

1) BAM: We append the BA module to concentrate the bands with more discriminative information based on KL divergence. The model BA + A1 + A2 outperforms A1 + A2 BA + A1 + A2 + AM is better than the model BA + A1 + A2, here the AM refers to the FSAM + FPAM module. It is observed that the OA with the BA module is more competitive than the network without it, especially for the KSC data. We can conclude that the BAM presented in this article is a very effective solution, especially for the data set with sparse sample distribution and classification difficulty.

To further evaluate the influence of the BA module, in this section, we compare the proposed BA block with

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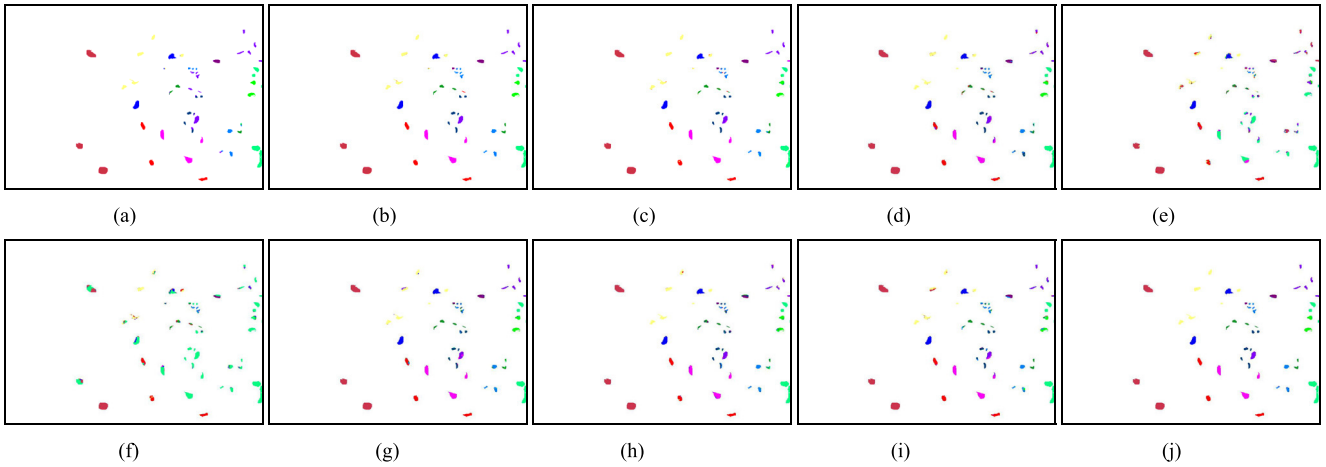


Fig. 9. Classification maps of the KSC with compared methods. (a) EPF. (b) ITCIMC. (c) MFASR. (d) CNNDH. (e) PCA3-F. (f) PCA5-F. (g) FADC. (h) Bi-RNN. (i) ResNet34. (j) FADCNN.

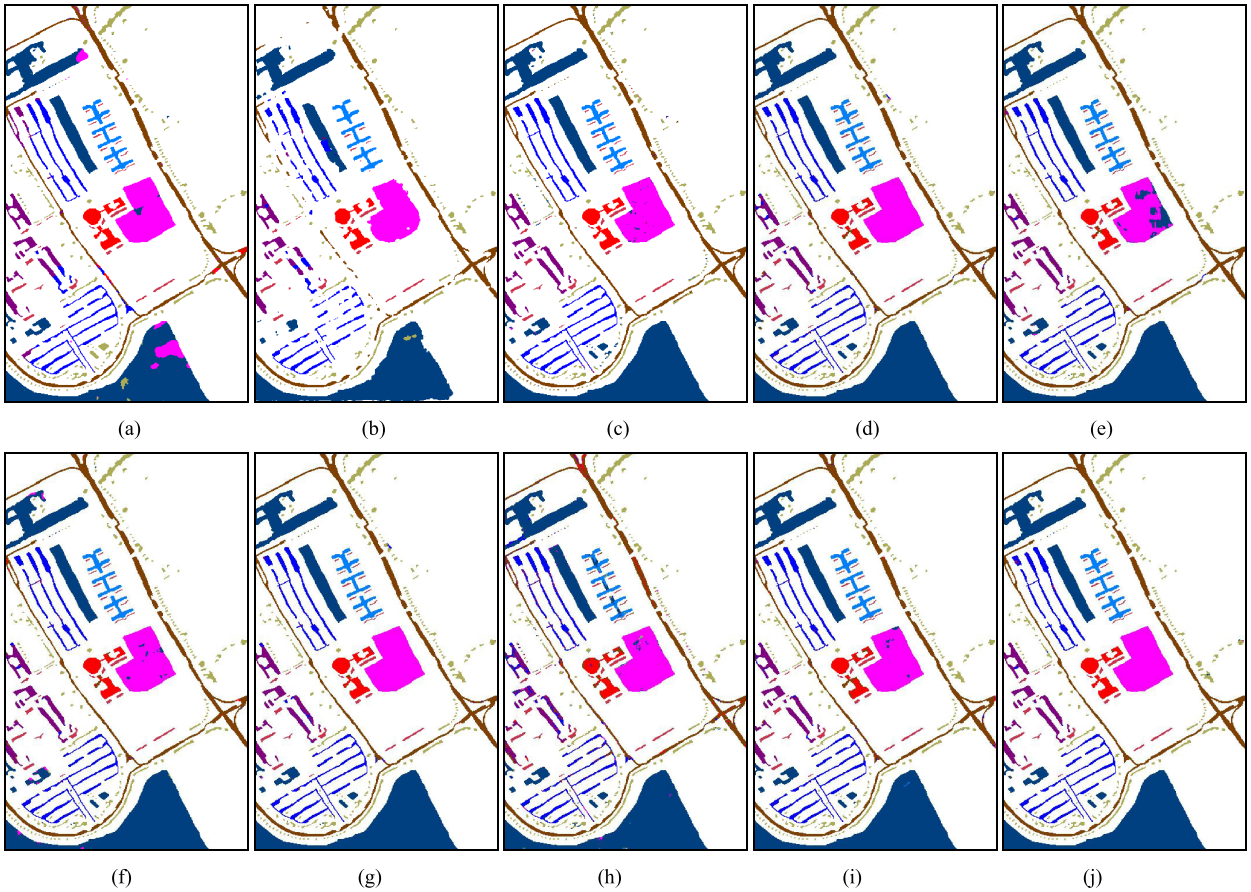


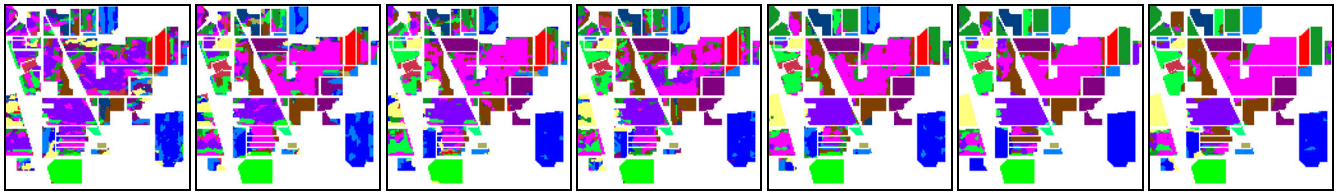
Fig. 10. Classification maps of the Pavia University with compared methods. (a) EPF. (b) ITCIMC. (c) MFASR. (d) CNNDH. (e) PCA3-F. (f) PCA5-F. (g) FADC. (h) Bi-RNN. (i) ResNet34. (j) FADCNN.

the PCA approach on the three data sets. As shown in Tables V, VI, and VII, it can be seen that the mechanism can improve better performance with less time consumption.

2) Attention Module (FSAM + FPAM): Based on the baseline network, we conduct a comparison experiment to detect the effect of the attention modules. Specifically, compared with the A1 + A2 results, the A1 + A2 + FSAM + FPAM produced a better performance for the three data sets. It demonstrated that the network with the attention modules generates better performance.

3) FM Module: Another experiment is conducted to compare the classification performance of the proposed architecture with the FM module. As shown in Table III, the improvements achieved by the model with FM block

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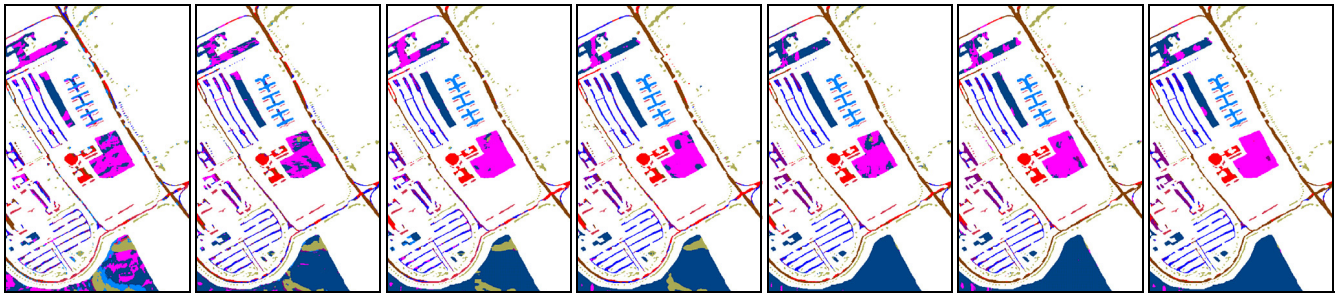
(a) (b) (c) (d) (e) (f) (g)

Fig. 11. Classification maps of the Purdue Indiana Indian Pines Scene with different sample number (K) of each class. (a) $K = 5$. (b) $K = 10$. (c) $K = 15$. (d) $K = 20$. (e) $K = 30$. (f) $K = 40$. (g) $K = 50$.



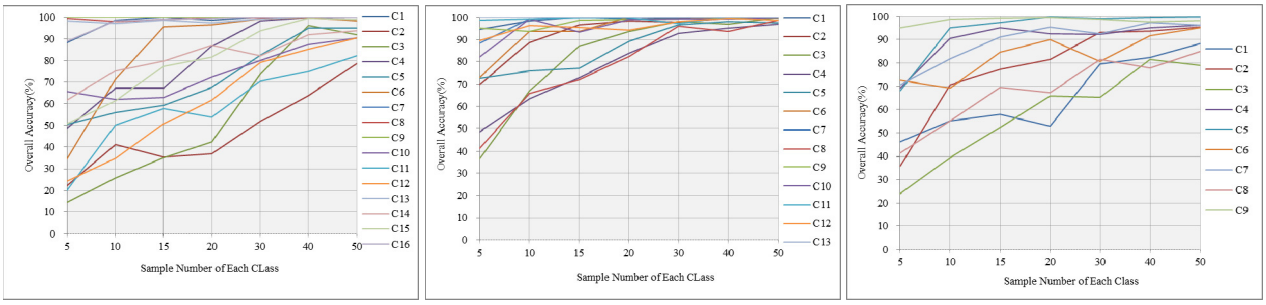
(a) (b) (c) (d) (e) (f) (g)

Fig. 12. Classification maps of the Salinas valley with different sample number (K) of each class. (a) $K = 5$. (b) $K = 10$. (c) $K = 15$. (d) $K = 20$. (e) $K = 30$. (f) $K = 40$. (g) $K = 50$.



(a) (b) (c) (d) (e) (f) (g)

Fig. 13. Classification maps of the University of Pavia with different sample number (K) of each class. (a) $K = 5$. (b) $K = 10$. (c) $K = 15$. (d) $K = 20$. (e) $K = 30$. (f) $K = 40$. (g) $K = 50$.



(a) (b) (c)

Fig. 14. Classification performance with a fixed number of each class for the proposed FADCNN. (a) Purdue Indiana Indian pines scene. (b) KSC. (c) University of Pavia.

demonstrate that the FM module is beneficial to the proposed model, aiming to improve the spatial feature in the framework to yield better performance.

To demonstrate the contribution of the FM module clearly, Table VIII is illustrated as four parts in different colors separately, the OAs provided with the same color are denoted as the similar models with or without FM. As can be seen, all the models with FM block are more effective than the ones without it. In specific, the OAs of the first two adjacent columns (A1 + A2 and A1 + A2 + FM) illustrate that the FM block is more significant for the simpler HSIC model. The OA improves 1.04% for Purdue, 2.36% for KSC, and 1.36% for Pavia. With more blocks participating in the feature extraction, the difference between the models with the FM module and the one without it becomes less noticeable. As recorded in green color, the OA enhances 0.11%, 0.08%, and 0.22% for Purdue, KSC, and Pavia, respectively. Meanwhile, it is worthy to note that the A1 + A2 + BA method produces more

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TABLE VI
OA FROM THE CLASSIFICATION RESULTS WITH ALL THE COMPARED METHODS OF THE KSC

Class P_{OA}%	EPF	ITCIMC	MFASR	CNNDH	PCA3-F	PCA5-F	FADC	Bi-RNN	ResNet34	FADCNN
1	**100.00**	95.65	**100.00**	98.13±2.29	99.18±0.30	97.45±4.52	99.21±1.04	99.04±0.60	98.42±0.92	99.97±0.06
2	83.68	96.01	96.58	72.84±5.90	43.13±4.77	43.95±7.51	86.99±4.11	72.71±4.92	70.71±15.99	**98.68±1.58**
3	72.17	96.99	**97.59**	95.39±3.61	4.85±0.71	13.05±20.25	85.78±4.14	69.93±7.03	86.70±8.47	96.72±2.27
4	**99.58**	98.73	89.67	65.95±3.86	22.46±2.36	37.70±14.32	75.56±6.85	67.61±2.09	66.46±4.32	92.86±2.33
5	94.20	89.44	**97.93**	82.24±4.64	50.43±1.72	56.03±17.39	88.57±3.00	63.34±7.53	67.49±5.60	97.14±1.36
6	99.59	97.12	**98.63**	86.20±7.09	10.39±2.54	22.62±23.52	88.30±5.15	85.08±1.52	71.25±18.61	97.73±1.09
7	96.43	**100.00**	96.00	92.96±4.02	14.86±1.73	23.43±26.98	95.24±4.42	99.38±0.93	87.99±15.53	99.24±0.79
8	**100.00**	98.78	**100.00**	87.19±3.35	9.47±5.45	18.00±23.60	91.78±2.70	76.94±13.24	88.18±5.83	98.98±0.69
9	95.00	**100.00**	94.44	92.08±0.76	5.69±6.73	20.15±38.94	97.35±1.34	91. ±3.9443	95.75±2.09	99.77±0.21
10	87.55	93.93	97.71	86.29±3.65	68.42±16.04	69.41±13.65	91.58±1.04	92.63±3.53	96.35±3.74	**100.00±0.00**
11	88.47	94.70	99.00	99.76±0.53	42.77±34.49	37.42±32.58	98.62±1.98	98.51±0.80	99.55±0.74	**99.86±0.32**
12	97.64	95.45	95.31	94.83±3.64	72.05±3.78	73.16±5.40	91.97±1.11	96.70±0.70	90.15±7.59	**100.00±0.00**
13	99.51	98.54	99.46	**100.00±0.00**	60.39±21.31	65.70±17.78	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
OA	90.78	95.09	97.83	91.86±0.93	47.93±8.61	52.58±15.12	93.95±0.56	89.97±2.11	90.94±1.68	**99.10±0.13**
AA	94.13	96.09	97.20	88.76±3.33	38.78±7.84	44.47±18.96	91.61±3.07	85.64±3.60	86.08±6.88	**98.53±0.82**

TABLE VII
OA FROM THE CLASSIFICATION RESULTS WITH ALL THE COMPARED METHODS OF THE UNIVERSITY OF PAVIA

Class P_{OA}%	EPF	ITCIMC	MFASR	CNNDH	PCA3-F	PCA5-F	FADC	Bi-RNN	ResNet34	FADCNN
1	**100.00**	95.65	**100.00**	99.02±0.30	98.32±0.94	98.65±1.19	98.27±1.54	91.40±0.63	96.40±0.88	99.70±0.25
2	83.68	96.01	96.58	**99.85±0.02**	99.39±0.94	99.21±0.38	99.80±0.24	97.66±1.01	99.78±0.12	99.75±0.14
3	72.17	96.99	**97.59**	91.79±0.70	94.04±5.88	88.89±9.42	93.20±1.78	73.51±12.24	95.84±0.68	94.31±3.16
4	**99.58**	98.73	89.67	99.23±0.08	98.39±2.39	99.00±0.18	98.23±0.47	89.74±5.17	98.95±0.36	99.10±0.37
5	94.20	89.44	97.93	99.73±0.37	99.94±0.13	99.24±1.69	**100.00±0.00**	83.75±8.76	99.55±0.18	99.82±0.25
6	99.59	97.12	98.63	99.93±0.08	91.81±9.07	97.04±2.63	**100.00±0.00**	94.94±2.16	98.7±0.80	99.92±0.07
7	96.43	**100.00**	96.00	92.36±0.74	93.23±3.59	93.55±0.98	97.85±1.69	79.40±11.42	86.85±7.32	96.97±2.53
8	**100.00**	98.78	**100.00**	98.17±0.14	98.97±0.83	98.92±1.06	97.93±1.54	84. 85±8.56	96.89±0.79	97.90±1.41
9	95.00	**100.00**	94.44	99.93±0.06	98.08±0.76	98.42±0.46	99.09±0.40	96.88±2.97	99.02±0.05	98.97±1.46
OA	90.78	95.09	97.83	98.91±0.04	97.76±1.27	98.13±1.02	98.92±0.35	92.37±2.18	98.19±0.05	**99.19±0.29**
AA	94.13	96.09	97.20	97.78±0.28	96.91±2.72	96.99±2.00	98.26±0.85	88.01±5.88	96.89±1.24	**98.49±1.07**

discriminative ability than the approach with A1 + A2 + FM modules.

In summary, we have two observations according to the result of each ablation: 1) the baseline network generates the worst performance, and the performance continuously increases with more modules employed and 2) the ablation analysis demonstrates the necessity of the BA module and the FM block.

E. Analysis of the Effect of Attention Blocks

To evaluate the contribution of the attention modules, we have conducted the HSIC experiments with different weights of the A1-CNN and A2-CNN for all the data sets. In this experiment, the weight coefficients are denoted as λ_1 for A1-CNN, and λ_2 for A2-CNN, respectively. Thus, (14) turns into

$$F = \sum_{l=1}^{L} (\lambda_1 \otimes F_1(X_l)) \circ (\lambda_2 \otimes F_2(X_l)) \tag{15}$$

where $\lambda_1 + \lambda_2 = 1$.

We have discussed the effectiveness of the weights changed by modifying the value of λ_1 and λ_2 starting from 0 to 1. All the comparison results of the three data sets are delivered in Table IX. It can be straightly observed that the best accuracies for the Pavia and KSC data sets are gained with the setting of $\lambda_1 = 0.5, \lambda_2 = 0.5$, and the best OA of the Purdue data is 98.03%, which is generated with the choice of $\lambda_1 = 0.3, \lambda_2 = 0.7$, respectively.

Furtherly, we extensively analyze the effect of the different pooling layer type of the proposed FASM and FPAM modules. We compute the informative map with the max-pooling and the average pooling layer jointly in the proposed attention modules. The max-pooling operation generates the salient map with the most considerable value in the local region. The average-pooling layer calculates the average value of the local area to represent the critical information of the feature map. In this section, we conducted a series of comparison experiments to clarify the strengths of the attention modules with different pooling layers. Specifically, the average-pooling layer and the max-pooling operation are equipped for FSAM and FPAM separately and alternatively. We report the

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TABLE VIII

ABLATION STUDY ON THE THREE HSI DATA SETS. WE REPORT THE OA VALUES (%) WITH VARIOUS MODULES IN 5000 ITERATIONS

Module Description	BA					√	√	√	√
	A1	√	√	√	√	√	√	√	√
	FSAM			√	√			√	√
	A2	√	√	√	√	√	√	√	√
	FPAM			√	√			√	√
	FM		√		√		√		√
HSI Data	Purdue	96.13	97.17	96.64	97.25	97.39	97.50	97.57	97.95
	KSC	91.17	93.53	93.45	93.95	98.65	98.73	98.91	99.1
	Pavia	97.35	98.71	98.08	98.92	98.33	98.55	98.52	99.17

TABLE IX

PERFORMANCE COMPARISON (%) OF THE PROPOSED FRAMEWORK WITH DIFFERENT COEFFICIENTS FOR A1-CNN AND A2-CNN

Coefficient Setting		OA		
λ_1	λ_2	Purdue	KSC	PAVIA
0.1	0.9	95.40	98.34	98.43
0.2	0.8	96.36	98.64	98.31
0.3	0.7	96.56	98.46	98.20
0.4	0.6	97.37	98.60	98.33
0.5	0.5	**97.95**	**99.10**	**99.19**
0.6	0.4	97.34	98.83	99.10
0.7	0.3	96.77	99.07	98.94
0.8	0.2	96.82	98.60	98.89
0.9	0.1	96.52	97.99	98.36

TABLE X

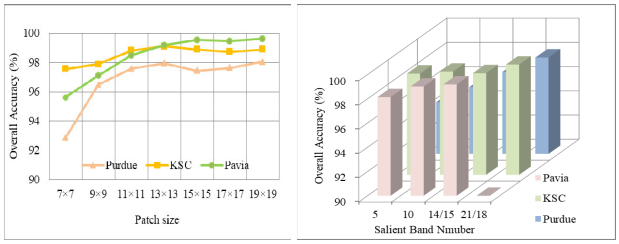
PERFORMANCE COMPARISON (%) OF THE PROPOSED MODEL WITH DIFFERENT TYPES OF POOLING LAYER. (A STANDS FOR AVERAGE, M STANDS FOR MAX)

Pooling Type	FSAM	A-Pooling	√	√	—	—	√
		M-Pooling	—	—	√	√	√
	FPAM	A-Pooling	√	—	√	√	√
		M-Pooling	—	√	—	—	√
HSI Data	Purdue		97.64	97.27	97.44	97.84	**97.95**
	KSC		98.96	98.7	98.58	98.7	**99.10**
	Pavia		98.97	98.85	98.55	98.94	**99.19**

classification results of all the three data sets with different choices in Table X. As can be observed, the best performances are obtained with the aggregation of the average-pooling and max-pooling layer, which indicates a general advantage of the mixed-up scheme for all the data sets. For other circumstances, the best accuracies are yielded with the same choice for the two attention blocks. The conduction with the max-pooling achieves OA of 97.84% for Purdue, and OAs of Pavia and KSC are 98.96% and 98.97 with the average-pooling operation.

F. Analysis of the Parameter Effect

In this section, the influence of three parameters on the performances of the FADCNN model are analyzed, including dropout rate, image patch size, and training sample percentage. Owing to the valuable spatial feature involved, the patch size of the training sample is crucial for the performance improvement of HSIC. Generally speaking, the bigger patch



(a) (b)

Fig. 15. (a) Effect of the patch size setting of the FADCNN. (b) Sample number. (b) Impact of the number of attention-based bands of the FADCNN.

size generates a higher performance. However, a larger size of the training sample caused the problem of the train-test data overlapping inevitably. Based on the extensive experimental results, we fixed the sample size as 13 × 13 empirically. On one side, we assume that the size setting is suitable to take advantage of the abundant local feature. On the other side, we consider this setting is not too large to lead to overlapping issues. Fig. 15(a) demonstrated qualitative comparisons of the classification performance with different patch sizes for all three data sets. To be more specific, the OA effectively improves with the patch size initially and decreases with the size getting larger for the KSC.

The relationship between patch size and OA concerning the training percentage toward three data sets is demonstrated in Fig. 16. As shown in this 3-D diagram, the vertical axis represents the classification accuracy OA, and the training ratio and the patch size is reflected with the horizontal axis and the third axis, respectively. It can be observed that the proposed model achieves the lowest OAs with a ratio of 1% and N of 7 for all the data sets. The overall accuracies mostly have a growth trend with a more prominent parameter, and the classification performance reaches the peak when the setting of most training ratio and the N of 13. Specifically, the OA gets the highest value with a portion of 10% and the size 13 × 13 for the Purdue Indiana Indian pines scene (97.95%). When the percentage is set to 10%, the best OA (99.10%) is achieved with N of 13 for the KSC, and the best performance (99.49%) is generated for the University of Pavia with the setting of $N = 15$ and the ratio of 9%, respectively. Overall, it indicates that the more spatial information implied in bigger patch size, which is beneficial for the improvement of the accuracy.

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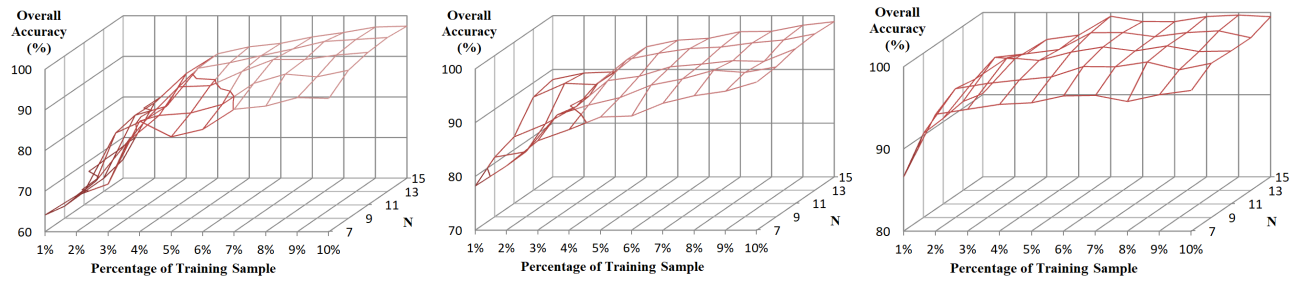


Fig. 16. Relationship between the training percentage and OA with different patch size of $N \times N$ for the proposed FADCNN. (a) Purdue Indiana Indian pines scene. (b) KSC. (c) University of Pavia.

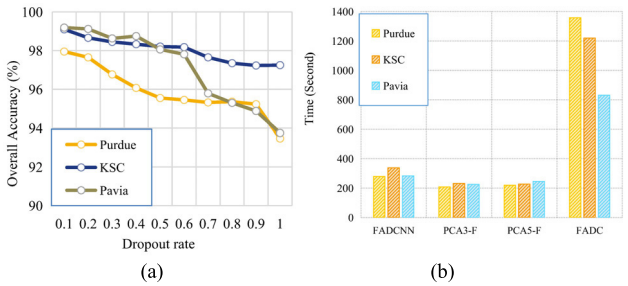


Fig. 17. Effect of the parameters of the FADCNN for the three data sets. (a) Impact of the dropout rate. (b) Time cost of the training process.

Fig. 17(a) demonstrates OAs obtained by the proposed network with different dropout rates on the three data sets. It can be observed that the highest OAs are yielded when the rate is 0.1, in specific, the best OA of the Purdue Indiana Indian pines scene of 97.95%, the KSC of 99.10%, and 99.19% for the University of Pavia. With the increase of the dropout rate, three curves have a decreasing trend similarly, in which the decline speed of the Purdue Indiana Indian pines scene is the fastest one. When the rate is set to 1, Purdue and Pavia generate the lowest OA of 93.47% and 93.75%, respectively, and the worst performance (97.23%) is yielded for the KSC with the rate of 0.9. Evidently, due to the active feature extracted from the attention-based dense network, the proposed model can generate the best performance with a lower dropout rate.

To evaluate the effectiveness of FADCNN with the number of the salient band, we completed the experiment with the different attention-based numbers for all the data sets. According to the number determined by the VD algorithm, the number varies from the set of {5, 10, 14} for Pavia, {5, 10, 15, 18} for Purdue, and {5, 10, 15, 21} for KSC, respectively. The obtained accuracy comparison is illustrated with colorful histograms in Fig. 15(b). As can be observed, the OAs are not satisfactory with a few salient bands, especially for the Purdue data with a value of 94.27%. With the more salient bands participated in the BA, the proposed HSIC model generates more preferable results.

Finally, the training time on the three data sets with compared methods is shown in Fig. 17(b). As shown in the histograms, on one side, we can find that the time cost is relatively similar in the FADCNN method and the PCA-based approach, while our BA-based way has better accuracy illustrated in Tables I, V, and VI. On the other side, compared to the proposed FADCNN model, the FADC model with full bands consumes more time due to the difficulty that existed in the training of the dense spectral blocks. In summary, it can be concluded that the proposed architecture is the most effective model with the best performance under the same parameters.

IV. Conclusion

In this article, an FADCNN approach for HSIC is proposed by using a dense connection of spatial and spectral information with a feedback attention mechanism. The proposed model enhances the transmission of the features with embedded dense blocks to concatenate the feature, and it effectively utilizes an attention map to improve the characterization of the extracted information. Besides, the FSAM and FPAM blocks are designed to strengthen semantic information with high-layer knowledge in a feedback style. Furthermore, we present a BAM to supply a strategy to balance the band weight for training, and an FM block is employed to explore the deep refinement of the incorporated spatial-spectral feature. Experiments and analysis illustrate that our model achieves excellent performance in terms of accuracy and computation on the public data sets.

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Chunyan Yu (Member, IEEE) received the B.Sc. and Ph.D. degrees in environment engineering from Dalian Maritime University, Dalian, China, in 2004 and 2012, respectively.

From September 2014 to September 2015, she was a Visiting Scholar with the College of Physicians and Surgeons, Columbia University, New York, NY, USA. She is an Associate Professor with the Information Science and Technology College, Dalian Maritime University. Her research interests include image segmentation, hyperspectral image classification, and pattern recognition.



Rui Han received the B.E. degree in software engineering from Ludong University, Yantai, China, in 2018. He is pursuing the M.A. degree in software engineering with Dalian Maritime University, Dalian, China.

His research interests include hyperspectral image processing and deep learning.



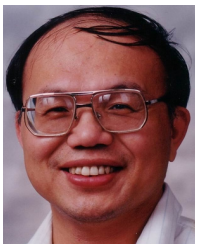
Meiping Song (Member, IEEE) received the Ph.D. degree from the College of Computer Science and Technology, Harbin Engineering University, Harbin, China, in 2006.

From 2013 to 2014, she was a Visiting Associate Research Scholar at the University of Maryland, Baltimore County, Baltimore, MD, USA. She is an Associate Professor with the College of Information Science and Technology, Dalian Maritime University. Her research includes remote sensing and hyperspectral image processing.



Caiyu Liu received the B.E. degree in network engineering from the Zhongyuan University of Technology, Zhengzhou, China, in 2019. She is pursuing the M.A. degree in computer science and technology with Dalian Maritime University, Dalian, China.

Her research interests include hyperspectral image processing and deep learning.



Chein-I Chang (Life Fellow, IEEE) received the B.S. degree in mathematics from Soochow University, Taipei, Taiwan, in 1973, the M.S. degree in mathematics from the Institute of Mathematics, National Tsing Hua University, Hsinchu, Taiwan, in 1975, the M.A. degree in mathematics from the State University of New York at Stony Brook, Stony Brook, NY, USA, in 1977, the M.S. and M.S.E.E. degrees from the University of Illinois at Urbana–Champaign, Urbana, IL, USA, in 1980 and 1982, respectively, and the Ph.D. degree in electrical engineering from the University of Maryland, College Park, College Park, MD, USA, in 1987.

He has been with the University of Maryland, Baltimore County (UMBC), Baltimore, MD, USA, since 1987 and is a Professor with the Department of Computer Science and Electrical Engineering. He is holding the Chang Jiang Scholar Chair Professorship and the Director of the Center for Hyperspectral Imaging in Remote Sensing (CHIRS), Dalian Maritime University, Dalian, China, since 2016, the Hua Shan Scholar Chair Professorship with Xidian University, Xian, China, since 2016, and the Feng-Tai Chair Professor with the National Yunlin University of Science and Technology since 2017. In addition, he has been also the Chair Professor with Providence University, Taichung, Taiwan, since 2012. He was a Visiting Research Specialist with the Institute of Information Engineering, National Cheng Kung University, Tainan, Taiwan, from 1994 to 1995, and also a Distinguished Visiting Fellow/Fellow Professor, both of which were sponsored by the National Science Council in Taiwan, from 2009 to 2010. Also, he was the Distinguished Lecturer Chair at National Chung Hsing University, Taichung, sponsored by the Ministry of Education in Taiwan from 2005 to 2006 and a Distinguished Chair Professor at National Chung Hsing University from 2014 to 2017. He has authored four books, *Hyperspectral Imaging: Techniques for Spectral Detection and Classification* (Kluwer Academic Publishers, 2003) and *Hyperspectral Data Processing: Algorithm Design and Analysis* (John Wiley & Sons, 2013), *Real Time Progressive Hyperspectral Image Processing*: *Endmember Finding and Anomaly Detection* (Springer, 2016), and *Recursive Hyperspectral Sample and Band Processing: Algorithm Architecture and Implementation* (Springer, 2017). In addition, he also edited two books, *Recent Advances in Hyperspectral Signal and Image Processing*, 2006 and *Hyperspectral Data Exploitation: Theory and Applications* (John Wiley & Sons, 2007) and co-edited with A. Plaza a book on *High Performance Computing in Remote Sensing*, CRC Press, 2007. His research interests include multispectral/hyperspectral image processing, automatic target recognition, and medical imaging.

Dr. Chang is a fellow of SPIE. He received the National Research Council Senior Research Associateship Award from 2002 to 2003 sponsored by the U.S. Army Soldier and Biological Chemical Command, Edgewood Chemical and Biological Center, Aberdeen Proving Ground, Maryland. He was a Plenary Speaker of the Society for Photo-optical Instrumentation Engineers (SPIE) Optics + Applications, Remote Sensing Symposium, in 2009. He was also a Keynote Speaker at the User Conference of Hyperspectral Imaging 2010, December 30, 2010, Industrial Technology Research Institute, Hsinchu; the 2009 Annual Meeting of the Radiological Society of the Republic of China, 2009, Taichung; the 2008 International Symposium on Spectral Sensing Research in 2008; and the Conference on Computer Vision, Graphics, and Image Processing 2003, Kimen and 2013, Nan-Tou, Taiwan. He holds seven patents on hyperspectral image processing. He was a Guest Editor of a Special Issue of the *Journal of High Speed Networks* on *Telemedicine and Applications* in April 2000 and a Co-Guest Editor of another Special Issue of the same journal on Broadband Multimedia Sensor Networks in Healthcare Applications in April 2007. He is also the Co-Guest Editor of Special Issues on High Performance Computing of Hyperspectral Imaging for the *International Journal of High Performance Computing Applications* in December 2007, Signal Processing and System Design in Health Care Applications for the *EURASIP Journal on Advances in Signal Processing* in 2009, Multipsectral, Hyperspectral, and Polarimetric Imaging Technology for the *Journal of Sensors* and *Hyperspectral Imaging and Applications* for *Remote Sensing* in 2017. He was an Associate Editor in the area of hyperspectral signal processing for IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING from 2001 to 2007. He is an Associate Editor of *Artificial Intelligence Research* and also on the Editorial Board of the *Journal of High Speed Networks*, *Recent Patents on Mechanical Engineering*, the *International Journal of Computational Sciences and Engineering*, the *Journal of Robotics and Artificial Intelligence*, and the *Open Remote Sensing Journal*.

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